The empirical recurrence for OEIS A304424, recovered from its tail

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Abstract

OEIS A304424 records "Empirical recurrence of order 90" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 159$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 94$. Its last coefficient is $c_{90} = 207 \neq 0$, so no further tail satisfies anything shorter; any order-90 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A304424 is "Number of nX5 0..1 arrays with every element unequal to 1, 2, 3, 4 or 5 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$3, 383, 5227, 98190, 1712347, 30362267, 536823135, 9501018264, \ldots$$

The entry states:

> Empirical recurrence of order 90 (see link above)

As of the "Last modified" line on the live entry (May 12 2018 09:37:07, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 159$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 159$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 90: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 90.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 318$ exact terms from $n = 4$ on, it returns order 90 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{90} c_i\, a(n-i),$$

c_1	9	c_{24}	519688251052	c_{47}	−214826071351166	c_{70}	−75939245508
c_2	202	c_{25}	−616363204120	c_{48}	−102944398472064	c_{71}	−259535928134
c_3	−2	c_{26}	62127190694	c_{49}	−28321594490037	c_{72}	−306531640944
c_4	−12548	c_{27}	−2330353088236	c_{50}	−30522074786114	c_{73}	−257399096564
c_5	−57830	c_{28}	−793242409987	c_{51}	−81186908951755	c_{74}	−108500082035
c_6	135274	c_{29}	1691476653815	c_{52}	−136041059300323	c_{75}	−10301808171
c_7	1890561	c_{30}	−6871496813353	c_{53}	−31976901526141	c_{76}	−23728509
c_8	4171293	c_{31}	−1911855268963	c_{54}	105408241335212	c_{77}	7195649581
c_9	−14355905	c_{32}	−17073250372563	c_{55}	144922087218401	c_{78}	8719253418
c_{10}	−90232531	c_{33}	−415571991664	c_{56}	53389435940114	c_{79}	4853996386
c_{11}	−118808986	c_{34}	4306654445326	c_{57}	−37863236609894	c_{80}	2409029428
c_{12}	361646934	c_{35}	−3462064337500	c_{58}	−75785069692932	c_{81}	814142016
c_{13}	1723371533	c_{36}	38863588733249	c_{59}	−23849194246496	c_{82}	227270509
c_{14}	2498629807	c_{37}	20831374257181	c_{60}	8274094452626	c_{83}	51177968
c_{15}	−2117625025	c_{38}	45053950840716	c_{61}	11839854819089	c_{84}	5798839
c_{16}	−14657547056	c_{39}	−11691202463397	c_{62}	516014006486	c_{85}	−133462
c_{17}	−26485033009	c_{40}	3024499255692	c_{63}	−10864254043647	c_{86}	−229680
c_{18}	−18576647204	c_{41}	62215519374725	c_{64}	−499102081606	c_{87}	179918
c_{19}	36725172430	c_{42}	76180148404791	c_{65}	4558756883895	c_{88}	18454
c_{20}	60549622786	c_{43}	189414020835603	c_{66}	4961332212423	c_{89}	5769
c_{21}	148875269022	c_{44}	65579002883136	c_{67}	3132387912629	c_{90}	207
c_{22}	112923072945	c_{45}	−39303927821814	c_{68}	567393988759		
c_{23}	229970672321	c_{46}	−50453398238038	c_{69}	71278613992		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 94$, and it is the recurrence OEIS A304424 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{90} - c_1 x^{90-1} - \cdots - c_{90}$. Its constant term is $-c_{90} = -207$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 90 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 90, so $\deg q = 90$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 90, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 90, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 207.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-90 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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